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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

Past Paper · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 60 / 60 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Functional-Skills-Maths-Level-2-Past-Paper-QP-A

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 1: Shopping centre

1 (a) Jessie is the manager of a shopping centre.

She knows that:

- $\frac{3}{5}$ of the shops sell clothes
- $\frac{1}{8}$ of the shops sell electronic goods
- all of the other shops sell food.

What fraction of the shops sell food?

[2 marks]

Q1(a) Worked Solution:

11
40
24 S
and seen OR
40 40
29
seen
[2 marks]
Tip: Award 2 marks if correct answer given

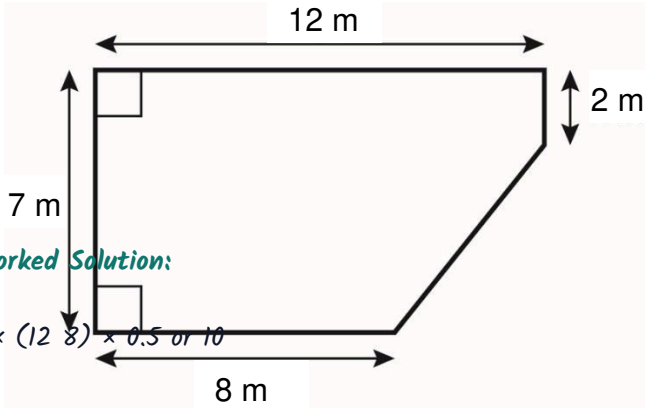
Your answer:

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Examiner use only

1 (b) Jessie will let this unit in the shopping centre:



Not drawn accurately

Q1(b) Worked Solution:

– –
(7 2) × (12 8) × 0.5 or 10
OR
–
(8 + 12) × 0.5 × (7 2) or 50

She has information about the floor areas of other available units in the shopping centre:

74 (m²)
67.6

Unit floor area (m^2)	51.1	82.3	67.6	49.8	71.2	40.7	59.4	68.9	90.2

Jessie thinks that the unit she will let has a floor area 12% larger than the median floor area of other available units.

Is she correct?

Show how you decide.

[5 marks]

Their 67.6 × 1.12 or 67.6 + 8.112 or 75.712
Their 74 ÷ 1.12 or 66.0(7..)
74 67.5 or 6.5
67.6 x 0.12 or 8.112
[5 marks]

Your answer:

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1 (c) Jessie lets a unit with a floor area of 40 m^2 for 12 months.

The unit costs £27.59 per m^2 per month.

What is the total cost of letting this unit for 12 months?

[2 marks]

Q1(c) Worked Solution:
(£)13243.20
4
27.59 × 40 × 12 or 13243.2
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: £

PAST PAPER

Please turn over

Examiner use only

1 (d) Jessie needs to work out how much bonus she will pay each of her workers for the last month.

She uses this formula to work out the bonus payment: $B = \frac{0.015T}{N}$

Where: B = bonus payment per worker (£)
T = total sales (£)
N = number of workers

Last month the total sales were £38 000

Jessie had 3 workers.

Work out the bonus payment each worker will get.

[3 marks]

Q1(d) Worked Solution:

(£)190
(0.015 × 38000) ÷ 3
570
[3 marks]
Tip: Award 3 marks if correct answer given

Your answer: £

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Examiner use only

1 (e) Jessie writes a report about the number of shoppers at the shopping centre.

She writes,

"Last week, we were only open for 6 days because of a bank holiday. The mean number of shoppers was 21 thousand a day and the range was 19.5 thousand shoppers."

Jessie has this information about this week:

Q1(e) Worked Solution:

23.2 or 23 200
27.1 or ~~27 100~~

Day of the week	Shoppers this week (in thousands)
Monday	15.4
Tuesday	12.5
Wednesday	17.3
Thursday	19.8
Friday	26.2
Saturday	39.6
Sunday	31.6

Compare this week and last week using the means and the ranges.

[3 marks]

Valid comparison of both mean and range
e.g. the mean number of shoppers this week was greater than last week, but the daily number of visitors was less consistent this week.
2: Water savings (Calculator
[3 marks]
Tip: Do not allow is

Your answer:

[Total marks: 15]

This is the end of Section A.

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Activity 2: Water savings

2 (a) Kamrul wants to reduce the amount of water he uses at home.

He uses the kitchen tap for 350 days a year for 20 minutes each day.

The flow rate of his kitchen tap is 5.5 litres per minute.

Kamrul wants a new kitchen tap with a flow rate of 4.7 litres per minute.

He thinks he can save more than 6000 litres of water in a year if he uses the new kitchen tap for 350 days a year for 20 minutes each day.

Is Kamrul correct?

Show how you decide.

[2 marks]

Q2(a) Worked Solution:

(5.5 4.7) × 20 × 350 or
(5.5 × 20 × 350) (4.7 × 20 × 350) or
38500 32900 or 5600
OR
6000 ÷ 350 or 17.14.. AND (5.5 4.7) × 20 or 16
No AND 5600 or
No AND 17(14..) and 16
[2 marks]

Your answer:

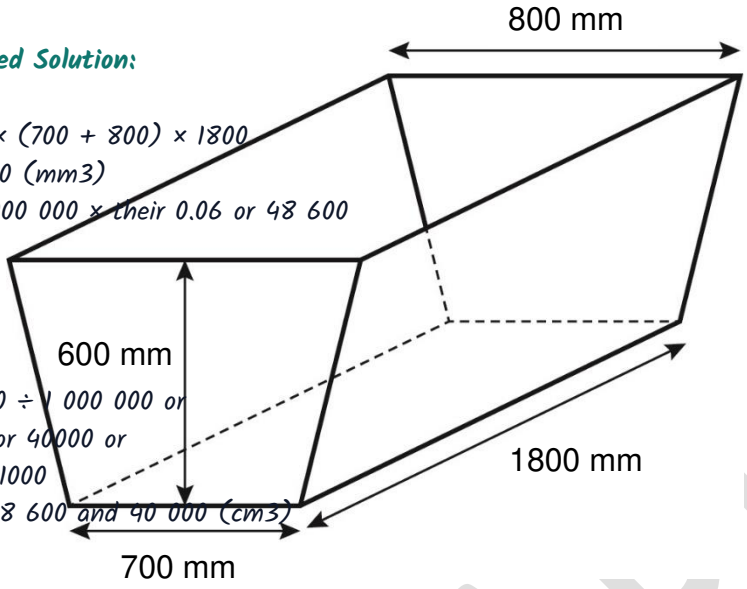
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Examiner use only

2 (b) The diagram shows Kamrul's bathtub.

The bathtub is a prism with a cross section in the shape of a trapezium.



Not drawn accurately

Q2(b) Worked Solution:

1

0.5 × 600 × (700 + 800) × 1800

810 000 000 (mm3)

Their 810 000 000 × their 0.06 or 48 600

5

000

48 600 000 ÷ 1 000 000 or

40 × 1000 or 40000 or

810 000 ÷ 1000

Yes AND 48 600 and 40 000 (cm3)

[5 marks]

Tip: Accept consistent units of mm^3 or cm^3

Kamrul uses this formula to calculate the capacity of this prism:

$$C = 0.5h(a + b)L$$

Where:

C = capacity of the prism
a = width of the bottom edge of the trapezium
b = width of the top edge of the trapezium
h = height of the prism
L = length of the prism

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Examiner use only

Kamrul normally fills the bathtub with water to 78% of its capacity.

He thinks that, if he fills the bathtub to 72% of its capacity, he will save more than 40 litres of water.

Is Kamrul correct?

Show how you decide.

[5 marks]

Your answer:

PAST PAPER

Please turn over

Examiner use only

2 (c) Kamrul wants to wash his car.

He knows it takes 120 litres of water to wash a car in a car wash.

Kamrul decides to hand wash his car.

He fills up a 3-gallon bucket six times to wash his car.

Q2(c) Worked Solution:

What percentage of 120 litres does Kamrul save?

Use the conversion: 1 gallon = 4.55 litres

[3 marks]

31.75(%)
3 × 4.55 or 13.65 or
3 × 6 × 4.55 or 81.9 or
120 ÷ 4.55 or 26.37..
–
[120 (3 × 6 × 4.55)] ÷ 120 × 100 or
(3 × 6) ÷ (120 ÷ 4.55) × 100
[3 marks]
Tip: Award 3 marks if correct answer given

Your answer: %

PAST PAPER

Examiner use only

2 (d) Kamrul looks at his quarterly water bill.

Total volume of water used this quarter: 118 m^3

This is 23% less than the same quarter last year.

Q2(d) Worked Solution:

Work out the volume of water Kamrul used for the same quarter last year.

Give your answer to 1 decimal place.

[2 marks]

153.2 (m3)

118 ÷ 0.77 or 153(.24..)

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer: **m^3**

PAST PAPER

Please turn over

Examiner use only

2 (e) The water company needs to reduce water leakage.

This data shows the volumes of water leaked in 2019:

Water leaked (millions of litres per day)	Number of days
$0 \leq w < 200$	87
$200 \leq w < 400$	105
$400 \leq w < 600$	153
$600 \leq w < 800$	20
Total	365

Q2(e) Worked Solution:

Use the data to work out an estimate of the mean amount of water leaked per day.

358 (millions of litres per day)

Give your answer to the nearest whole number.

[3 marks]

(87 × 100) + (105 × 300) + (153 × 500) + (20 × 700) or
8700 + 31500 + 76500 + 14000 or
130700
358(.082..)
6
festival (Calculator Test)
[3 marks]
Tip: Award 3 marks if correct answer given

Your answer: **millions of litres per day**

[Total marks: 15]

PAST PAPER

Examiner use only

Activity 3: Music festival

3 (a) Pavel organises a music festival.

He puts this information on the festival's website:

"Four million, nine hundred and eight thousand, six hundred and thirty people attended our festival over the last 30 years".

Write four million, nine hundred and eight thousand, six hundred and thirty as a number.

[1 mark]

Q3(a) Worked Solution:

4 908 630

[1 mark]

Your answer:

Tip: Accept commas or spaces

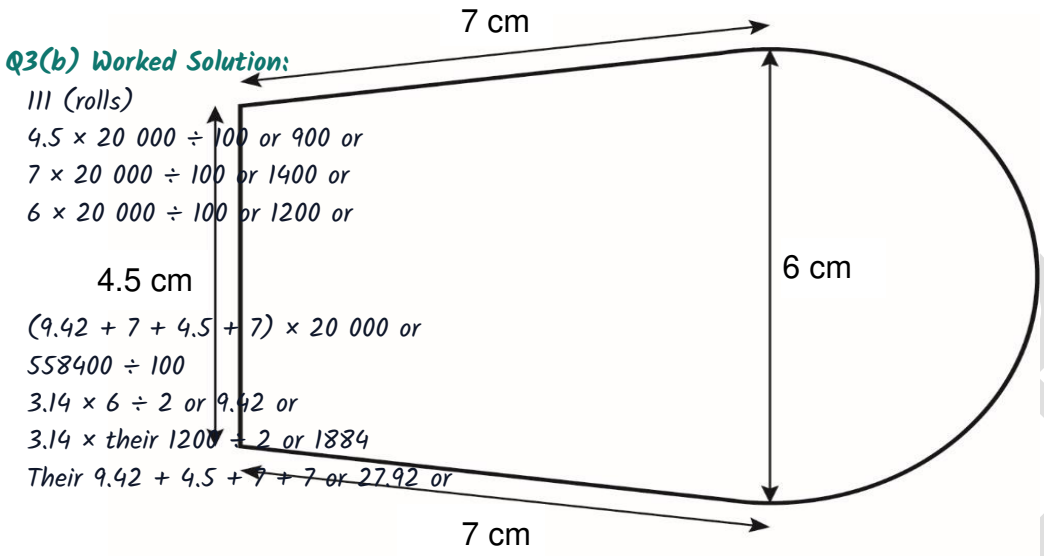
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Examiner use only

3 (b) Pavel has this scale drawing of the festival area.

It is made up of a trapezium and a semicircle.



Scale 1 : 20 000

He needs to order fencing to go around the festival area.

[5 marks]

He will leave gaps totalling 65 m for the gates.

Fencing comes in rolls 50 m in length.

Tip: Award 5 marks if correct answer given

Q3(b) Worked Solution:

111 (rolls)

4.5 × 20 000 ÷ 100 or 900 or

7 × 20 000 ÷ 100 or 1400 or

6 × 20 000 ÷ 100 or 1200 or

(9.42 + 7 + 4.5 + 7) × 20 000 or

558400 ÷ 100

3.14 × 6 ÷ 2 or 9.42 or

3.14 × their 1200 ÷ 2 or 1884

Their 9.42 + 4.5 + 7 + 7 or 27.92 or

their 1884 + their 900 + 2 × their 1400 or

5584

or 5585

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Examiner use only

How many rolls does Pavel need for the festival area?

Use $\pi = 3.14$

[5 marks]

Your answer: **rolls**

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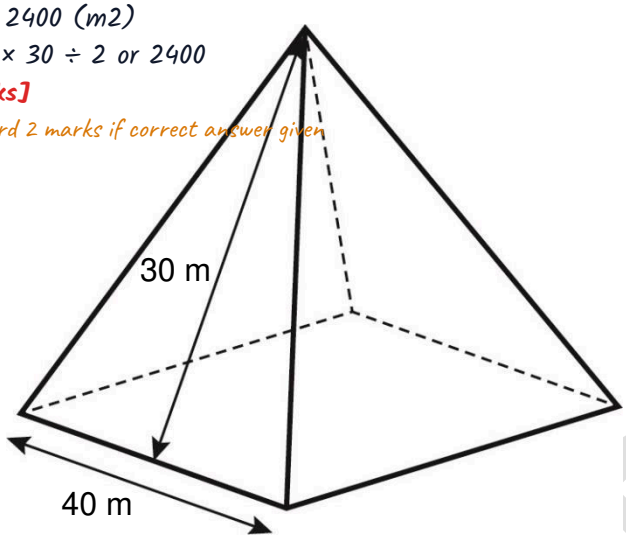
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Examiner use only

3 (c) The main stage is a square-based pyramid.

Pavel needs to work out the surface area of the four triangular faces of the pyramid.

Q3(c) Worked Solution:
No and 2400 (m2)
4 × 40 × 30 ÷ 2 or 2400
[2 marks]
Tip: Award 2 marks if correct answer given



Not drawn accurately

He uses this formula: $A = 4\left(\frac{bh}{2}\right)$

Where:

A = surface area of the 4 triangular faces of the pyramid (m^2)
b = length of the base of the triangular face (m)
h = slant height of the triangular face (m)

Pavel thinks the total surface area of the 4 triangular faces of this pyramid is $1200\,m^2$

Is Pavel correct?

Show how you decide.

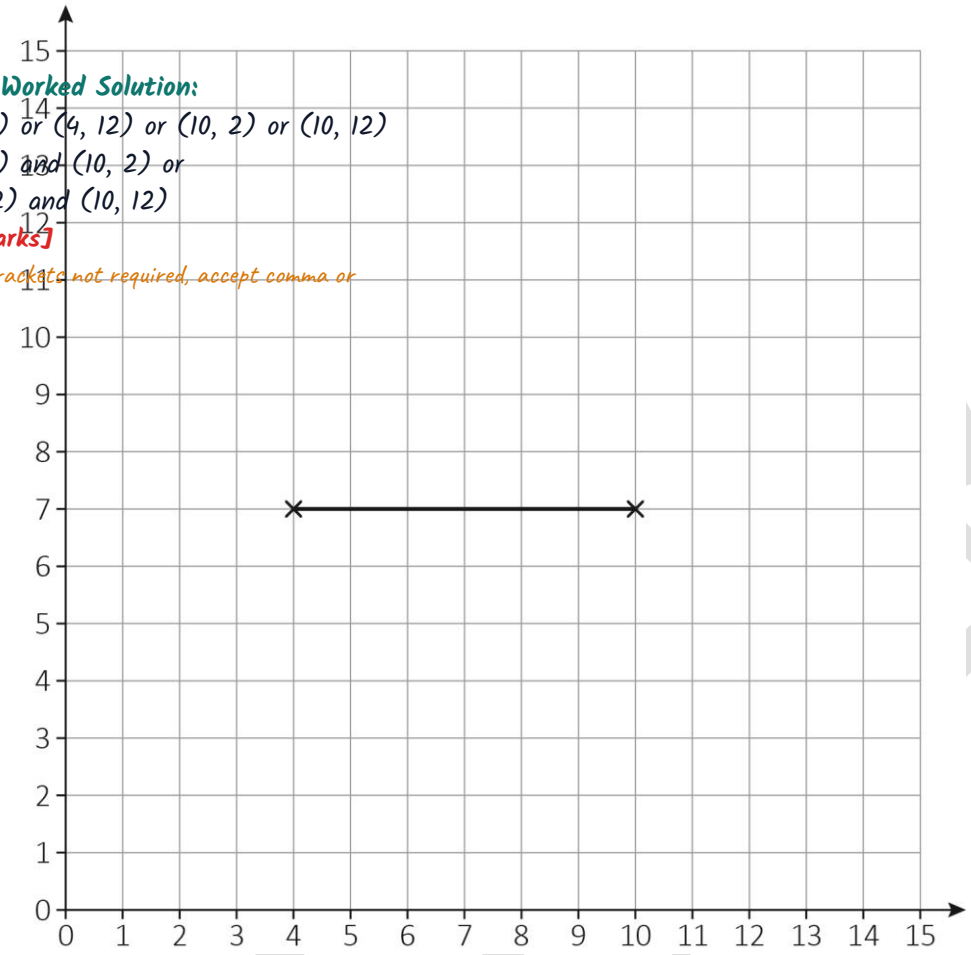
[2 marks]

Your answer:

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Examiner use only

3 (d) The coordinate grid shows the space for the car park:



Q3(d) Worked Solution:
(4, 2) or (4, 12) or (10, 2) or (10, 12)
(4, 2) and (10, 2) or
(4, 12) and (10, 12)
[2 marks]
Tip: Brackets not required, accept comma or

The car park is rectangular. It measures 120 m by 100 m

Each square on the grid represents 20 m by 20 m

Pavel draws two vertices and one edge of the car park area on the grid.

Write down possible coordinates for the other two vertices of the car park.

[2 marks]

Your answer:

Please turn over

Examiner use only

3 (e) The area where the festivalgoers will have their tents is 314 acres.

Pavel knows that for safety reasons each tent requires 27 m^2

He thinks that more than 50 000 tents can fit into this area.

Is he correct?

Q3(e) Worked Solution:

7

1

Show how you decide.

Use the conversion: 1 acre = 4046 m^2

[3 marks]

314 × 4046 or 1 270 444

Their 1 270 444 ÷ 27 or 47 053.48..

OR

Their 1 270 444 ÷ 50 000 or 25.4(...)

No AND 47 053(.48..) (tents)

[3 marks]

Tip: Accept No and 1 270 444 and 1 350 000

Your answer:

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Examiner use only

3 (f) The probability that a festivalgoer chosen at random has travelled by car is $\frac{14}{25}$

What is the probability that they **did not** travel by car?

Give your answer as a decimal.

[2 marks]

Q3(f) ~~Worked Solution:~~
0.44
– 14 11
1 or or 0.56
25 25
admissions (Calculator Test)
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer:

[Total marks: 15]

PAST PAPER

Please turn over

Examiner use only

Activity 4: College admissions

4 (a) Mel is an apprentice in a college admissions office.

She is organising interviews for new applicants.

The ratio of interviewers to applicants each day should be no higher than 2 : 15

There are 5 interviewers available on Monday.

Q4(a) Worked Solution:

37 What is the maximum number of applicants Mel should invite on Monday?

5 ÷ 2 × 15 or 37.5

[2 marks]

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer:

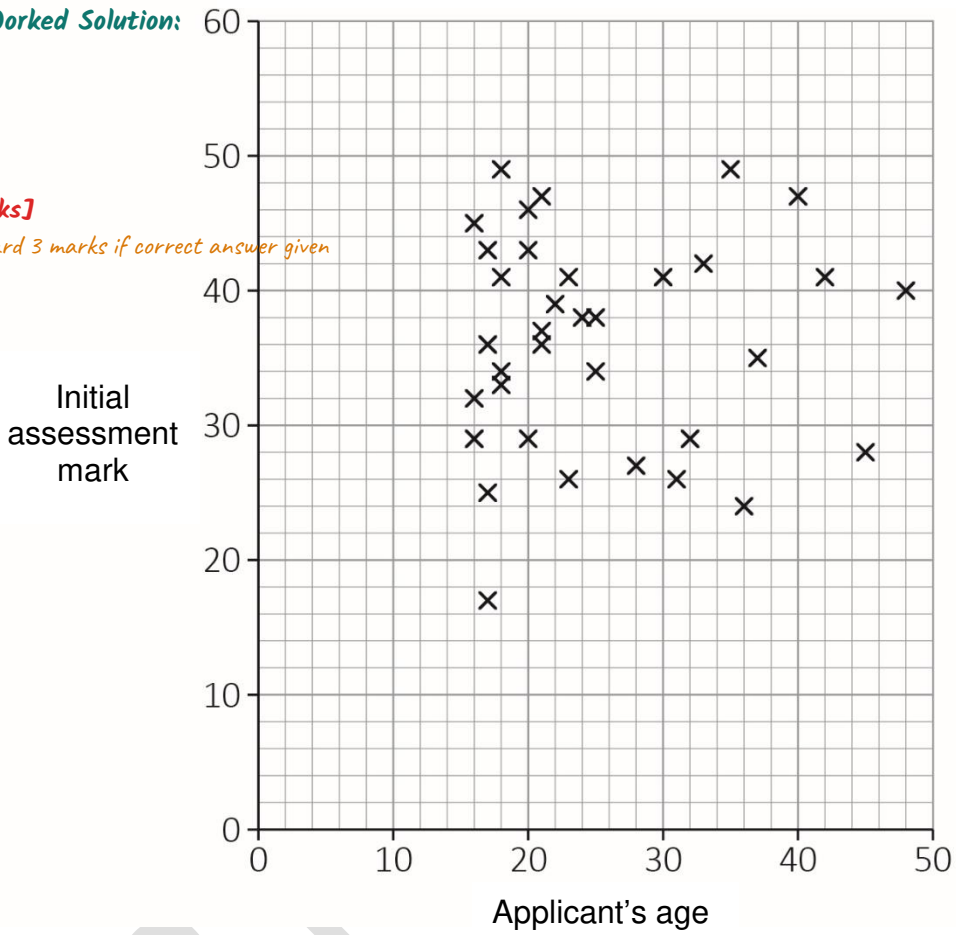
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Examiner use only

4 (b) All applicants who attend the interview complete an initial assessment.

Mel has this scatter diagram about the results of the initial assessment:

Initial assessment marks and age of applicants



Q4(b) Worked Solution:
5
12
8
for:
[3 marks]
Tip: Award 3 marks if correct answer given

What fraction of applicants over the age of 26 scored more than 40 marks?

[3 marks]

Your answer:

Please turn over

Examiner use only

4 (c) Mel has information about applicants accepted on two new courses:

	Course A	Course B
Male applicant	22	71
Female applicant	59	48

Q4(c) Worked Solution:

What is the probability one of these applicants chosen at random is female and accepted on Course B?

Give your answer as a percentage.

[2 marks]

24(%)
48
OE or 0.24
200
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: ____ %

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4 (d) Mel needs to write a report about the income from tuition fees.

She has this information about tuition fees:

Course	Fee
A	£390
B	£650
C	£720
D	£460
E	£390
F	£980
G	£720
H	£980
J	£720

Mel writes in her report,

'The college has a £1.7 million income target this year.'

If 1476 students enrol onto courses and pay the modal tuition fee, the college will achieve more than $\frac{5}{8}$ of this target.

Is this statement correct?

Show how you decide.

[4 marks]

Your answer:

Please turn over

PAST PAPER

Question 4(d) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q4(d) Worked Solution:

1
(£)720
1 700 000 ÷ 8 × 5 or 1 062 500
Their 1 062 500 ÷ their 720 or 1475.69..
or
Their 1 062 500 ÷ 1476 or 719.85..
Yes AND 1475.6(9..) or
Yes AND (£)719(.85..) and (£)720

[4 marks]

Tip: Allow full method to find of any value

Examiner use only

4 (e) The conversion rate is the proportion of all applicants who go on to enrol on a course.

This proportion is expressed as a decimal.

Mel needs to compare the conversion rates for this year and last year.

Q4(e) Worked Solution:

Mel has this information about enrolments this year:

- all applicants: 7680
- applicants who enrolled on a course: 4167

Last year, 6 out of every 11 applicants enrolled on a course.

Is the conversion rate this year lower than the conversion rate last year?

Show how you decide.

[2 marks]

6 ÷ 11 or 0.5454.. or
4167 ÷ 7680 or 0.5425.. OR
6 ÷ 11 × 7680 or 4189.09.. OR
4167 × 11 ÷ 6 or 7639.5
Yes AND 0.545(4..) and 0.542(5..) OR
Yes AND 54.5(4..%) and 54.2(5..%) OR
Yes AND 4189(.09..) OR
Yes AND 7639(.5.)
[2 marks]
Tip: Award if 54.2(5...) rounded 54.3

Your answer:

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Examiner use only

4 (f) A new student takes out a student loan of £1450 for 2 years at a compound rate of 3% per annum.

Work out the amount of interest owed on this loan after 2 years.

[2 marks]

Q4(f) ~~Worked Solution:~~

(£)88.30 or (£)88.31

–

1450 × 1.032 1450 or 88.305

10

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer: £

[Total marks: 15]

This is the end of the assessment.

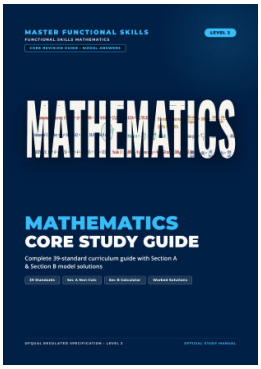
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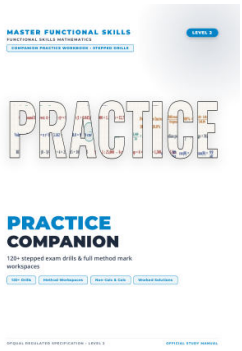


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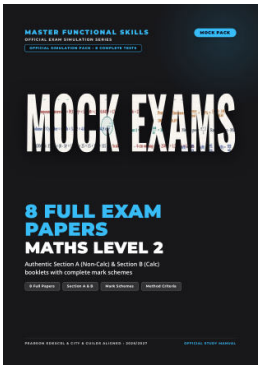


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